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ORIGINAL ARTICLE

Association Between Social Networks and Prehospital Delay in Care of Patients With Acute Ischemic Stroke in China

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BACKGROUND: Prompt hospital arrival is critical for acute ischemic stroke (AIS) treatment. This study aimed to examine the association between social network structure and prehospital delay among patients with AIS in China.

METHODS: A cross-sectional survey using a validated instrument was conducted among patients with AIS admitted to 2 general hospitals in Qingdao, China, between October 2020 and May 2021. Key inclusion criteria were confirmed AIS diagnosis, age ≥18 years, and ability to complete the survey. Exclusion criteria included severe cognitive impairment or aphasia. We assessed 5 social network characteristics using egocentric network methodology: effective network size (number of nonredundant contacts), kin proportion (percentage of relatives), betweenness centrality (role as a bridge in the network), network constraint (limitations in forming new connections), and network efficiency (ability to access resources through contacts). Prehospital delay was defined as a binary outcome (arrival to hospital >3 hours after symptom onset). Binary logistic regression was used to analyze associations between network features and delay, adjusting for clinical and demographic variables.

RESULTS: Among 422 participants (mean age 65.0±11.7 years, 71.0% male), 270 (64.0%) experienced prehospital delay (>3 hours). Adjusted logistic regression revealed that while a larger effective network size (odds ratio [OR], 0.39 [95% CI, 0.24–0.64]) and a higher kin proportion (OR, 0.21 [95% CI, 0.05–0.78]) were associated with reduced odds of delay, higher betweenness centrality (OR, 1.16 [95% CI, 1.03–1.31]), greater network constraint (OR, 1.02 [95% CI, 1.00–1.03]), and higher network efficiency (OR, 1.04 [95% CI, 1.02–1.06]) were associated with increased odds.

CONCLUSIONS: Social network structure significantly influences prehospital delay in patients with AIS. Interventions focused on social network expansion and strengthening kin involvement may reduce delays and improve stroke outcomes.

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Key Words: China ■ emergency medical services ■ ischemic stroke ■ patients ■ social networking

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The World Health Organization identifies stroke as 1 of the 3 leading causes of death worldwide,[1] and it is the leading cause of mortality in China, with acute ischemic stroke (AIS) accounting for ≈85% of cases.[2] AIS is characterized by a high incidence rate, significant morbidity, and considerable disability.[3] In China, there are over 70 million individuals with stroke, and ≈2 million new cases are diagnosed each year, reflecting an annual increase of 8.7% in incidence.[4] Intravenous administration of r-tPA (recombinant tissue-type plasminogen activator) is one of the most effective treatments for AIS.[5] Evidence-based research indicates that timely intravenous thrombolysis within the therapeutic window is an effective early intervention for patients with AIS.[6,7] Despite the availability of effective therapies, prehospital delay remains the main reason for the low thrombolysis rates among these patients.[8] Therefore, implementing effective management strategies to reduce prehospital delays is a critical research priority.

In recent years, numerous studies have focused on identifying patient characteristics and risk factors contributing to prehospital delays, such as sex, age, living

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WHAT IS KNOWN

- Rapid arrival at a hospital after a stroke is critical for patients to receive effective treatment.

WHAT THE STUDY ADDS

- This study confirms a link between social network structure and delayed hospital arrival in patients with acute ischemic stroke.
- It provides practical insights for medical staff to develop effective interventions aimed at reducing prehospital delays by considering patients' social networks.

Nonstandard Abbreviations and Acronyms

AIS	acute ischemic stroke
OR	odds ratio
r-tPA	recombinant tissue-type plasminogen activator

conditions, family history of stroke, education level, timing of symptom onset, symptom severity, symptom recognition, and perceived severity.[9–14] Despite these findings, few studies have explored how individual interactions influence prehospital delays.

Studies have demonstrated that key risk factors for prehospital delay include lack of social support and insufficient disease knowledge among patients and their family members.[9,15,16] The significant role of social networks in shaping patients' health behaviors has attracted increasing attention from researchers.[17] Social networks refer to the web of relationships connecting individuals within their social environments.[18] Numerous epidemiological studies have confirmed that individual health behaviors can be influenced by the structure of social networks.[19,20] Research demonstrates that patients' social networks can serve as social capital, facilitating access to emergency medical services and playing a crucial role in symptom recognition and healthcare needs.[10] The onset of a stroke is often a collective event involving at least 1 caregiver, along with family members, friends, or even strangers, who communicate to determine whether and when to seek medical help.[10] Therefore, delays are frequently the outcome of communication, decision-making, and coordinated actions among several witnesses.[21]

However, few studies have examined prehospital delays from a social network perspective. Boden-Albala et al[22] and Madsen et al[18] suggested that strong relationships provide patients with stroke with greater support, promote earlier symptom recognition, encourage timely medical treatment, and ultimately reduce prehospital delays. Conversely, Dhand et al[10] suggested that patients surrounded by family members often exist within closed networks, which may create a false consensus. This can lead them to adopt a wait and see approach when symptoms occur, resulting in delays. The role of intimacy in prehospital delay remains controversial. Therefore, we aimed to systematically analyze the social network characteristics of patients with AIS through social network analysis to clarify how these characteristics influence prehospital delay.

METHODS

Study Design and Participants

The data that support the findings of this study are available from the corresponding author on reasonable request. The study was conducted in 4 medical wards across 2 university hospitals: The Affiliated Hospital of Qingdao University and Qingdao Municipal Hospital. Both institutions are part of a university hospital network. A cross-sectional design was used, and patients were recruited between October 2020 and May 2021. Stroke symptom onset was defined as the moment when either the patient or a bystander first noticed the symptoms.[9] Prehospital delay was defined as the time interval from symptom onset to hospital arrival, with a delay classified as present if this duration exceeded 3 hours.[9] Therefore, the primary outcome variable was binary, categorizing patients as arriving within ≤ 3 hours or > 3 hours of stroke symptom onset. This classification was based on a combination of medical records, reviews of acute treatments administered, and patient reports. The 3-hour threshold was selected because thrombolysis is generally recommended within 4.5 hours of stroke onset, and the median in-hospital delay in China exceeds 1 hour.[6] In addition, the 3-hour cutoff has been widely used in previous studies on prehospital delay.[9,15,16]

The final sample for this study included 422 patients, of whom 270 experienced prehospital delays. Patients were eligible for inclusion if they met the following criteria: (1) aged 18 years or older; (2) diagnosed with AIS based on clinical examination and neuroimaging (including at least 1 computed tomography or magnetic resonance imaging brain scan); and (3) willing and able to participate in the study. Exclusion criteria were as follows: (1) diagnosis of a transient ischemic attack or cerebral hemorrhage; (2) symptom onset occurring while already admitted to a hospital or nursing home (as these patients do not experience prehospital delay); (3) cognitive impairment or an inability to answer interview questions; and (4) undetermined time of symptom onset or hospital admission. This study was conducted in accordance with the Declaration of Helsinki and was reviewed and approved by the Medical Ethics Committee of The Affiliated Hospital of Qingdao University. The ethical approval number is QYFYWZLL25899.

The researcher first reviewed the medical records of selected patients to identify those who met the inclusion and exclusion criteria. After obtaining informed consent, the purpose and significance of the study were explained to each participant, and the questionnaire was administered. The self-report questionnaire was either completed independently by the patients or read aloud to them by the investigator on an item-by-item basis. Patients were encouraged to make their

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own decisions, and the investigator recorded each response verbatim. Because most of the study participants were elderly, a verbal survey method was used to enhance data reliability and accuracy. In this approach, each item was read aloud to the participant, who then dictated their response while the investigator filled out the questionnaire on their behalf. To mitigate patient recall bias, we not only inquired about the time of stroke onset but also conducted detailed interviews regarding the specific manifestations of initial symptoms. A neurologist was involved to assist in determining the true time of onset. Furthermore, our investigators performed in-depth interviews with both patients and their family members to reconstruct the timeline as accurately as possible. This study exclusively enrolled patients who provided written informed consent, and in accordance with protocols prioritizing participant autonomy, no data were collected from eligible individuals who declined to participate. To minimize nonresponse bias, several strategies were used: face-to-face interviews were conducted at flexible times outside treatment periods to reduce burden, and it was explicitly communicated that participation was voluntary and would not affect medical care.

After data collection, validated questionnaire responses were structured into a matrix format using Microsoft Excel, and the matrices were subsequently analyzed using the University of California, Irvine NETwork software. The data were sorted and saved into folders based on the unique identification numbers assigned to each participant.

Measurements

We used an egocentric network analysis approach, in which each participant (referred to as the ego) identified their 10 closest social associates (or alters), including friends, family members, current or former romantic partners, and coworkers. A standardized social network survey was administered, following the 3 recommended steps outlined by Burt: name generator, name interrelater, and name interpreter.[23] A study coordinator conducted the social network survey at the patient's bedside, typically within 1 week of stroke hospitalization. Social networks were constructed based on the following questions (Figure 1A):

1. Who have you talked to the most in the past 3 months? When you experience personal issues (eg, work, family, or emotions), whom do you usually talk to? Please list their nicknames or last names (eg, Zhang).
2. With whom have you socialized in the past 3 months? (eg, visiting, traveling, dining, or square dancing).
3. Who reminded you to visit the hospital this time? Who accompanied you? If the participant repeats names, clarify by asking: "Are [Name 1] and [Name 2] the same person?" If fewer than 5 names are provided, prompt with: "Is there anyone else?" or "Whom would you call if you needed money?"
4. How close do you feel to each person you mentioned? (0=unfamiliar, 1=familiar). Ask for each listed individual.
5. Now, I will ask about the relationships between the people you mentioned. For example, between "Name 1" and "Name 2": Are they familiar with each other? (0=unfamiliar, 1=familiar).
6. Besides the people you already mentioned, is there anyone else who helped you during this event?

The social network data collected in this study primarily consists of 3 components: matrix data (Figure 1A), name-interpretation relationship data (Figure 1B), and name-interpretation feature data (Figure 1C). The overall data processing methodology is illustrated in Figure 1. The matrix data were analyzed using the University of California, Irvine NETwork software to extract network structural indicators, including network size, network density, and network constraint degree. The name-interpretation relationship data were collected by asking how long the patient had known each nominee, the frequency of interaction, and the degree of emotional closeness to assess the strength of these ties. The name-interpretation feature data were based on 4 key social attributes: age, marital status, educational level, and income level. The edit distance method was used to calculate the similarities of these attributes between the interviewee and each nominee, providing a measure of social attribute similarity within each participant's network.

Questionnaire on Influencing Factors of Medical Treatment

In alignment with the study's research objectives, the investigators designed a structured questionnaire to assess risk factors influencing the medical treatment process. The development of the questionnaire was informed by a comprehensive review of the relevant literature. It consisted of the following key components: (1) demographic and clinical characteristics, such as sex, age, past medical history, and primary symptoms; (2) sociological factors, including educational level, household income, and medical insurance type; (3) cognitive and behavioral factors affecting medical decision-making, such as patients' and family members' understanding of stroke, recognition of early warning signs, and perception of symptom severity; (4) transportation-related factors, including the use of emergency services (eg, dialing 120), referrals from the community or lower-level hospitals, and delays due to traffic congestion; and (5) time-related factors, such as the time of symptom recognition, duration of travel from home to the hospital, referral delays, traffic-related delays, emergency response time, and the interval from hospital arrival to treatment initiation. For further details, refer to Table 1.

Stroke Symptom Alert Assessment Questionnaire

In this study, Zhao et al[24] developed the Stroke Symptom Alert Assessment Questionnaire to evaluate the level of alertness among patients experiencing premonitory stroke symptoms. The primary aim of the questionnaire was to assess the awareness and recognition of early stroke signs among the general public or specific subgroups. It is designed to help researchers and medical professionals better understand individuals' ability to recognize stroke symptoms, their perception of stroke-related risks, and their responses when symptoms occur. This assessment tool can identify knowledge gaps and guide improvements in stroke education and prevention, ultimately enhancing public responsiveness to stroke warning signs. The questionnaire contains 9 items, each scored using a binary system: correct responses receive 1 point, while incorrect responses receive zero. The overall difficulty level of the questionnaire was moderate, with an average item difficulty (*P* value) of 0.77. The average discriminant index (D) was

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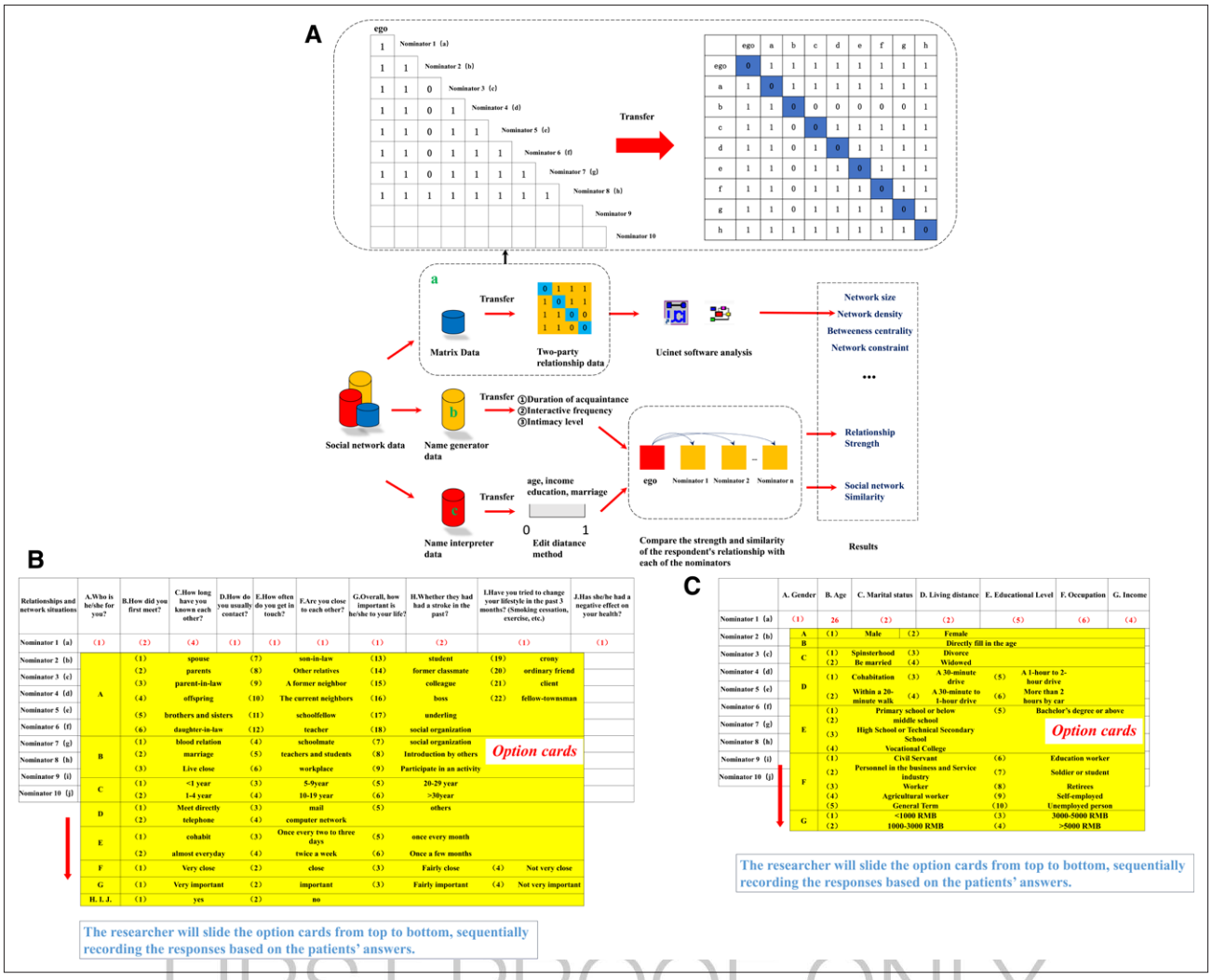

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Figure 1. Flowchart of social network data analysis process.
The social network data collected in this study comprises 3 main components: matrix data (name generator), name interrelater, and name interpreter. The analytical procedure is summarized in the figure. Matrix data (name generator) were processed using the University of California, Irvine NETwork software to quantify structural network characteristics such as network size, centrality, and constraint (**A**). Name interrelater data were evaluated through 3 dimensions—duration of acquaintance, interaction frequency, and emotional closeness—to assess the strength of the relationship between the respondent and each nominator, ultimately yielding an average measure of relational strength across the network (**B**). Name interpreter data pertaining to 4 key sociodemographic variables—age, marital status, education level, and income—were analyzed using the edit distance algorithm to compute pairwise similarity between the respondent and each nominator, resulting in network-level indices of similarity for each attribute (**C**).

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0.53, indicating good differentiation between high- and low-performing respondents. The content validity index was 0.84, suggesting the strong relevance of the items to the measured construct. The questionnaire also demonstrated high reliability, with a Kuder-Richardson 20 coefficient of 0.798 and a test-retest intraclass correlation coefficient of 0.729, indicating good temporal stability. For further details, refer to Table 2.

Statistical Analysis

We compared demographic and clinical characteristics between early and delayed arrival using an unpaired 2-tailed *t* test or Wilcoxon rank-sum test for continuous variables, depending on the distribution. For categorical variables, we applied the χ^2 (χ^2) test and the Fisher exact test when expected cell counts were fewer than 5. Bivariate logistic regression analyses were conducted to explore the relationships between arrival time (categorized as ≤3 hours or >3 hours) and various demographic, clinical, and social network characteristics. Subsequently, variables with a $P<0.1$ in the univariate analyses were eligible for inclusion in the multivariable analysis. All eligible variables were initially entered into a full multivariable logistic regression model. A stepwise backward elimination procedure was then applied to identify the final set of independent factors associated with delayed arrival, using a significance level of $P<0.05$ for retention in the model, and all analyses were performed at a 95% confidence level. The assumption of linearity for the continuous covariates in the logit was assessed using the Box-Tidwell procedure. Given that 19 variables were tested, a Bonferroni correction was applied, setting a stringent significance level of

Table 1. The Definition of Social Network Analysis Variables

Social network analysis variables	Definition
Effective network size	The individual social network size of patients with AIS is determined by the total number of peers each patient nominates.
Network density	The network density of an individual's social network reflects how interconnected the alters are with one another.
Mean degree	The mean degree of the individual social network of patients with AIS refers to the average number of connections each nonpatient member in the network has. It reflects the overall distribution of contact relationships within the network.
Degree centrality	It is a commonly used metric in social network analysis to assess the importance or influence of individual nodes within a network. The degree centrality of patients with AIS refers to the number of direct connections between patients and other individuals in their personal network. A higher number of direct connections indicates greater centrality.
Betweenness centrality	This metric reflects the extent to which a network member functions as a crucial, strategic mediator or bridge between other members. The greater the dependence of those in the surrounding network on the patient, the higher the patient's betweenness centrality.
Constraint	The constraint refers to the extent to which each network member is connected to other members within the network.
Efficiency	Network efficiency refers to the number of redundant connections within a network.
Network hierarchy	Hierarchy in a network refers to the degree to which relationships among members are asymmetrical in terms of reachability. A higher hierarchy score suggests a more rigid network structure and greater dominance of certain members.
Network similarity	To evaluate the homogeneity within each patient's social network, 4 key social attributes–age, marital status, and educational and income level–were collected from both patients and their nominated contacts. These attributes were used to compute network similarity indicators.
Relationship strength	This study assessed the strength of social network relationships using 3 key dimensions: duration of acquaintance, frequency of interaction, and degree of intimacy.
Percentage kin	The proportion of the number of people nominated by the patient who are relatives of the patient.

AIS indicates acute ischemic stroke.

α=0.0026 for the interaction terms. The results indicated no significant deviations from linearity (all P>0.0026), thus supporting the model specification with unmodified continuous variables.

Social Network Analysis

Details of the social network analysis variables are presented in Table 1; formulas for their calculation are provided in Supplementary File 1.

RESULTS

Sample Statistics

The sample characteristics are presented in Table 2. A total of 422 patients were enrolled in this study. Of these, 270 arrived at the hospital more than 3 hours after symptom onset, whereas 152 arrived within 3 hours. In the univariate analysis, no significant differences were observed between delayed and timely arrival in terms of demographic or clinical characteristics. However, notable differences were observed in several cognitive-behavioral and logistical factors. Specifically, patients who lacked knowledge about stroke (P<0.001), perceived their symptoms as nonsevere (P<0.001), or had lower stroke symptom alert scores (P<0.001) were more likely to arrive late. Living conditions also played a role: patients living alone were more likely to be delayed (P=0.007). In terms of transport-related factors, patients who did not arrive by emergency medical services (P<0.001), encountered traffic congestion (P=0.019), or lived farther from the hospital (P=0.046) experienced longer delays.

Compositional Social Network Characteristics

As shown in Table 3, 422 patients with stroke nominated a total of 2471 individuals within their social networks. Patients who experienced delayed hospital arrival were characterized by lower values in several social network metrics, including degree centrality, network hierarchy, effective network size, mean degree, network efficiency, marital status similarity, and income level similarity. However, they had a higher proportion of kin among their network members. In contrast, patients who arrived at the hospital within 3 hours exhibited 3 notable social network features: more people in their networks provided medical assistance, recognized stroke symptoms immediately, and promptly sought medical help. No significant differences were observed between the 2 groups regarding network constraint, betweenness centrality, density, relationship strength, age similarity, or educational level similarity. Logistic regression analysis was conducted to identify independent risk and protective factors associated with prehospital delays. The results remained consistent after adjusting for known confounders (Figure 2). According to bivariate logistic regression results (Figure 2), the following were identified as protective factors against delay: effective network size (odds ratio [OR], 0.39 [95% CI, 0.24–0.64]; P<0.001), percentage of kin in the network (OR, 0.21 [95% CI, 0.05–0.78]; P=0.02). Conversely, the following were found to be risk factors for prehospital delay: betweenness centrality (OR, 1.16 [95% CI, 1.03–1.31]; P=0.015), network constraint (OR, 1.02 [95% CI, 1.00–1.03]; P=0.014), and network efficiency (OR, 1.04 [95% CI, 1.02–1.06]; P<0.001).

DISCUSSION

This study demonstrated that a larger effective network size among patients with stroke was associated with a reduced likelihood of delays in seeking medical

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Table 2. Baseline Characteristics of Patients, According to Arrival Time (n=422)

Characteristic	Overall (n=422)	<3 h (n=152)	>3 h (n=270)	*P* value*
Male sex, n (%)	300 (71.09)	113 (74.34)	187 (69.26)	0.269
Age ≥65, y, n (%)	229 (54.26)	85 (55.92)	144 (53.33)	0.608
Living in the countryside, n (%)	134 (31.75)	41 (26.97)	93 (34.44)	0.114
Manual worker, n (%)†	265 (62.80)	102 (24.17)	163 (60.37)	0.148
Sudden onset	275 (65.17)	131 (87.5)	144 (53.33)	<0.001
Language impairment	197 (46.68)	84 (55.26)	113(41.85)	0.005
Family history of stroke, n (%)	95 (22.51)	40 (26.31)	55 (20.37)	0.160
Living alone, n (%)	34 (8.05)	5 (3.29)	29 (10.74)	0.007
Primary school and below, n (%)	120 (28.43)	40 (26.32)	80 (29.63)	0.525
Having medical insurance, n (%)	295 (69.90)	113 (74.34)	182 (67.40)	0.136
Arrived by EMS, n (%)	83 (19.67)	54 (35.53)	29 (10.74)	<0.001
Experienced transfer, n (%)	61 (14.45)	23 (15.13)	38 (14.07)	0.767
Encountered traffic jam, n (%)	150 (35.55)	43 (28.29)	107 (39.63)	0.019
No knowledge about stroke, n (%)	248 (58.77)	64 (42.11)	184 (68.15)	<0.001
Far away from the hospital, n (%)‡	110 (26.07)	31 (20.39)	79 (29.26)	0.046
History of prior stroke, n (%)	123 (29.15)	50 (32.89)	73 (27.04)	0.204
Onset during holidays, n (%)	148 (35.07)	54 (35.53)	94 (34.81)	0.883
Onset at night, n (%)	98 (23.22)	30 (19.74)	68 (25.19)	0.203
The patient feels that the symptoms are severe, n (%)	170 (40.28)	122 (80.26)	48 (17.78)	<0.001
Type of stroke, n (%)				
Large-artery atherosclerosis	217 (51.42)	79 (51.97)	138 (51.11)	0.893
Cardioembolic stroke	14 (3.32)	6 (3.95)	8 (2.96)	
Small vessel occlusion	152 (36.02)	52 (34.21)	100 (37.04)	
Stroke of other determined cause	0 (0.00)	0 (0.00)	0 (0.00)	
Unexplained stroke	39 (9.24)	15 (9.87)	24 (8.89)	
Ischemic stroke severity, n (%)§				
Mild (NIHSS score <6)	357 (84.60)	129 (84.87)	228 (84.44)	0.232
Moderate (NIHSS score 6–15)	57 (13.51)	18 (11.84)	39 (14.44)	
Severe (NIHSS score >15)	8 (1.90)	5 (3.29)	3 (1.11)	
Changes in symptoms before admission, n (%)				
No changes	87 (20.62)	28 (18.42)	59 (21.85)	0.724
Elevate	259 (61.37)	93 (61.18)	166 (61.48)	
Alleviate	46 (10.90)	19 (12.5)	27 (10.0)	
Fluctuate	30 (7.10)	12 (7.89)	18 (6.67)	
Stroke aura symptoms alert score, mean (SD)‖		5.47 (2.61)	3.21 (2.29)	<0.001

EMS indicates emergency medical services; and NIHSS, National Institutes of Health Stroke Scale.

*P value calculated from unpaired 2-tailed t test or Wilcoxon rank-sum test for continuous variables and χ^2 test and Fisher's exact test for categorical variables.

†Manual worker includes those engaged in construction, farming/forestry/fishing and related, installation and related, manufacture and production, transportation, and driver occupations.

‡Far from the hospital means the distance between the patient's onset and the final hospital is >20 km.

§NIHSS is a measure of stroke severity ranging from 0 (mild) to 42 (severe).

‖Stroke aura symptoms alert questionnaire-9 is a test for stroke aura symptoms alert.

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assistance. Patients with extensive and effective social networks can leverage these connections as medical capital to access a wide range of medical information and resources. The transition from univariate to multivariable analysis allows us to clarify the independent roles of social network factors by accounting for their intercorrelations. The observed shifts in variable estimates on adjustment reflect the process of isolating direct effects from the influence of confounding variables.[25,26] Specifically, betweenness centrality emerged as an independent risk factor only after adjustment, indicating that its effect was previously masked by correlated protective factors, such as emergency medical services use. Conversely, the protective effect of high income similarity

Table 3. Network Characteristics of Patients <3 and >3 Hours, Unadjusted

Variable	<3 h (n=152)	>3 h (n=270)	*P* value*
Network structure characteristics,† median [IQR]			
Degree centrality	7.00 [4.25–10.00]	5.00 [2.00–9.00]	<0.001
Constraint	38.93 [35.43–59.13]	40.94 [32.45–76.56]	0.537
Hierarchy	0.01 [0.00–0.02]	0.01 [0.00–0.05]	0.041
Effective size	1.80 [1.00–3.00]	1.00 [1.00–2.67]	0.020
Mean degree	4.37 [2.40–7.20]	2.62 [1.00–5.00]	<0.001
Betweenness centrality	0.45 [0.00–5.12]	0.00 [0.00–3.63]	0.128
Efficiency	0.33 [0.20–0.50]	0.50 [0.25–0.62]	<0.001
Density	0.91 [0.65–1.00]	1.00 [0.64–1.00]	0.205
Network composition characteristics‡			
Percentage of kin relationships, median [IQR]	30.00 [16.67–100.00]	42.86 [20.00–100.00]	0.005
Relationship strength, median [IQR]	26.37 [24.20–28.00]	26.80 [23.95–28.75]	0.254
High age similar, n (%)	69 (45.39)	122 (45.18)	0.967
High marriage similar, n (%)	136 (89.47)	220 (81.48)	0.030
High education similar, n (%)	39 (25.66)	73 (27.04)	0.758
High income similar, n (%)	40 (26.32)	48 (17.78)	0.038
No. of people who provide medical advice, median [IQR]	2.00 [1.00–3.00]	1.00 [1.00–2.00]	0.001
People around the patient at the time of illness, n (%)			
Family members	133 (87.5)	253 (93.70)	0.007
colleague	10 (6.58)	3 (1.11)	
Stranger or alone	9 (5.92)	14 (5.19)	
People around immediately recognize stroke at the first time, n (%)	86 (56.58)	69 (25.56)	<0.001
People around immediately seek medical help when the patient has symptoms, n (%)	144 (94.74)	173 (64.07)	<0.001

IQR indicates interquartile range.

*P value calculated from unpaired 2-tailed t test or Wilcoxon rank-sum test for continuous variables and χ^2 test and Fisher exact test for categorical variables.

†Network structure is a quantitative description of the arrangement of social ties in each patient's network. See definitions of each term in Methods.

‡Network composition is the range of characteristics of people around the patient. See definitions of each term in Methods.

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was attenuated and became nonsignificant, suggesting that its benefits are largely realized through other related pathways within the network.[27] Most notably, the reversal in the association of kin proportion—from a risk factor in univariate analysis to a strong independent protective factor (OR, 0.21)—strongly indicates the presence of underlying confounding. This pattern aligns with the Simpson paradox,[28] where aggregated analysis can obscure true relationships. By statistically balancing all included covariates, the multivariable model successfully isolated the genuine protective effect of kin support.[25,29] Collectively, these findings underscore that multivariable regression is essential for deciphering the complex and interdependent nature of social networks in prehospital delay.

Notably, our study also found that patients with highly efficient social networks were more likely to experience delays in seeking medical care. This counterintuitive finding can be explained by the Braess paradox, which describes situations in which adding resources to a network can actually decrease overall efficiency.[30] Originally observed in traffic networks, this paradox is also evident in the medical care–seeking networks of patients with stroke, especially when the network is highly efficient. In this context, increased resources—such as additional emergency care facilities, more convenient transportation routes, and abundant medical information—can paradoxically complicate the decision-making process for patients and their families, leading to delays. Although these additional options may seem beneficial, they can lead to more complex and prolonged decision-making. For example, (1) increased decision options: when faced with multiple choices for medical care, patients and their families may spend more time weighing the pros and cons, thus extending the decision-making process. A patient might hesitate between choosing the nearest emergency facility or one that is farther away but better equipped. (2) Impact of social networks: If individuals within the patient's social network fail to recognize stroke symptoms or understand

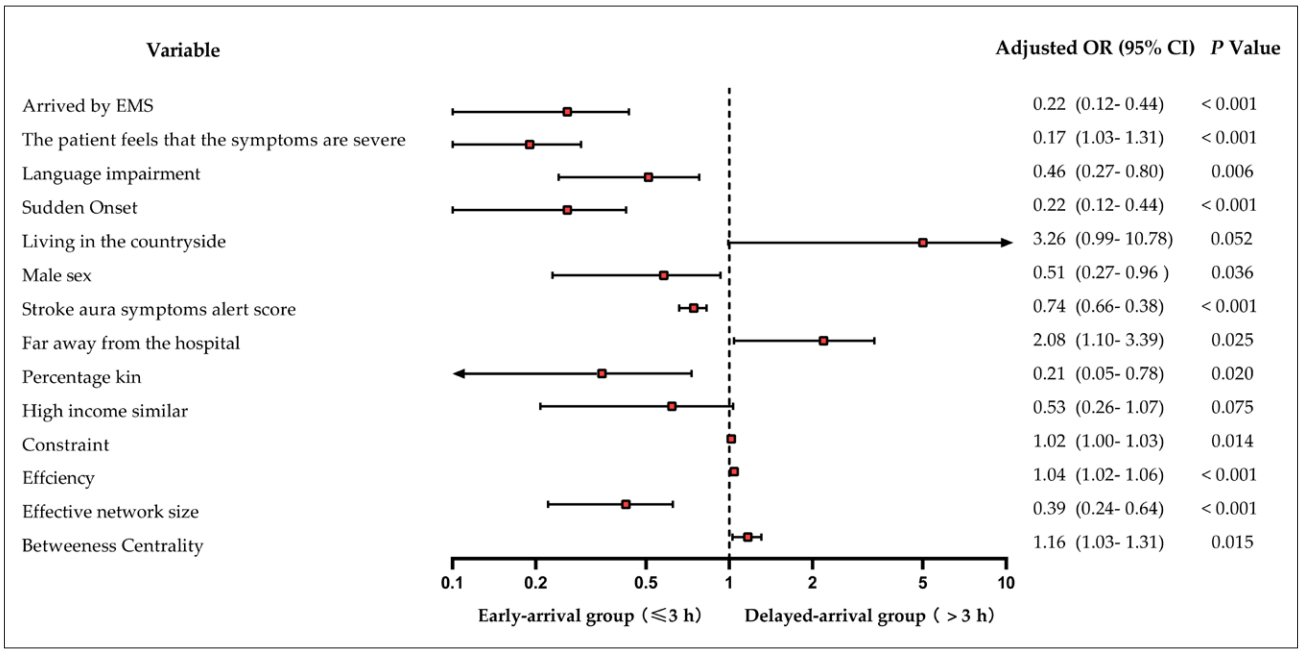

Figure 2. Multivariable analysis of determinants for prehospital delay after acute ischemic stroke.
Forest plot shows adjusted odds ratios and 95% CIs from the final multivariable logistic regression model (developed using backward stepwise selection) for prehospital delay (>3 hours vs ≤3 hours). The vertical line represents the null effect (odds ratio [OR], 1). Odds ratios are presented on a logarithmic scale. The complete numerical results for all variables in this final model are available in Table S1. EMS indicates emergency medical services.

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their severity, this can cause indecision about seeking medical care. For example, a patient may delay seeking immediate medical attention based on advice from friends or family members who do not appreciate the urgency of the situation.[31] (3) Information overload: An excess of available resources can result in information overload, causing patients to spend more time sorting through and understanding conflicting information. This can delay decision-making. For example, a patient might struggle to make quick decisions when faced with multiple conflicting medical recommendations.[32]

Patients with high network constraints—characterized by small social circles and closely connected groups—are more likely to experience delays. This finding aligns with the results reported by Dhand et al[10] In addition, some researchers have noted that living alone is associated with an increased risk of prehospital delay.[33] Patients with high social network constraints tend to operate within closed social networks, limiting access to diverse information and resulting in reliance on a single perspective. Conversely, a higher proportion of relatives within a patient's social network is associated with a reduced likelihood of delays. Patients who exhibit symptoms in the presence of others are 8 times more likely to arrive at the hospital within 3 hours than are those who live alone.[34] Dhand et al[10] suggested that in Western contexts, intimate family social networks are a primary factor contributing to delays in patients with stroke. They emphasized that kinship plays a crucial role in placing patients with stroke patients into tightly knit networks, which may increase the likelihood of delays. However, this study found different results in the Chinese context, likely due to cultural differences in family structures between Chinese and Western societies. American families tend to be nuclear, emphasizing individual privacy and independence. Medical decisions are typically made by immediate family members or designated proxies, reflecting a higher degree of social network constraint. In contrast, multigenerational households are common in China, where the presence of multiple family members results in lower social network constraints. This can facilitate quicker symptom recognition and prompt action, thereby reducing delays in seeking medical care.[35]

The findings of this study align with those of most previous studies, indicating that the rapid recognition of stroke symptoms and prompt action by those around the patient can reduce prehospital delays.[36–40] This study also found that patients with more intermediary care centers were more likely to experience delays. Specifically, individuals who are more dependent on others tend to ignore their symptoms, which consequently delays their hospital arrival. When Madsen et al[18] examined the association between sex and the relationship linking social networks to prehospital delay, they revealed that women living with their spouses were more likely to experience delays in accessing hospital care. They suggested that this may be due to women often assuming caregiver roles within their families, where visiting the emergency department could interfere with their ongoing

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responsibilities, thereby contributing to the delay. Similarly, Barr et al[41] reported that women are more likely to experience delays than are men. Building on this premise, the current study further quantitatively validates why dependent groups, such as caregivers, are more prone to delays, particularly due to the influence of social network structures. Previous research has noted that women's familial social networks often lack reciprocity, resulting in fewer benefits than those received by the individuals they live with. Even when a woman has a partner who encourages medical treatment, she may still be hesitant to seek help for stroke symptoms. Conversely, men are more likely to promptly access medical care once urged by their partners.[18]

Our findings emphasize the significance of social networks in stroke timely response. These results indicate that the structure and dynamics of social networks can significantly influence the timeliness and effectiveness of medical care-seeking behaviors among individuals experiencing stroke symptoms. Based on our findings, several strategies can be proposed to reduce prehospital delay among stroke patients: (1) Promote social participation: Encourage high-risk individuals to engage in social activities and expand their interpersonal interactions. This study shows that a larger and more effective social network is associated with reduced prehospital delays. Social relationships are essential for accessing stroke-related medical knowledge and emergency resources. Previous research has also indicated that patients with smaller social networks tend to have less awareness of stroke symptoms and appropriate responses.[42] Therefore, patients should expand their social circles by participating in activities, such as family gatherings and group exercises. (2) Enhanced stroke education: Expand public education efforts to focus on stroke prevention and symptom recognition. Our study found that 68.01% of participants received stroke-related information from medical professionals. Providing structured information during hospitalization and at discharge is a cost-effective strategy for improving awareness and preparedness.[33] Hospitals and community health centers should organize stroke-related educational activities for patients with a history of cardiovascular disease. Social groups can be established to regularly invite these individuals to training sessions covering early stroke symptoms, thrombolysis time windows, and basic first-aid measures.[43] In addition, online platforms can be used to share educational content and instructional videos, encouraging patients to disseminate this information within their social networks.[44] (3) Identifying and screening high-risk social networks: Community health workers should focus on individuals with a history of transient ischemic attacks or stroke. By conducting brief surveys to map patients' social networks, high-risk groups for prehospital delays can be identified. These include patients with high betweenness centrality, high network constraints, or high network efficiency. Population-based social network interventions can leverage interpersonal influences to significantly modify health behaviors.[45] For example, involving members of a patient's social network in health education programs has been shown to improve stroke preparedness.[46] Community caregivers can integrate family involvement with broader community engagement to identify high-risk individuals early through social network screening. Targeted stroke education and awareness campaigns can then be provided to expand social support for high-risk individuals and encourage them to seek and accept timely assistance from others.[47] (4) Integrating high-risk individuals into family support systems: Beyond patients' own awareness and decision-making, family and social support play a critical role in stroke response and care.[48] Family members are often the first to recognize stroke symptoms and are typically the ones who initiate emergency medical calls.[49] Strengthening stroke preparedness within families can provide emotional support for patients, encourage timely medical attention, and improve treatment adherence and outcomes.[50] Community health departments should advocate for the integration of high-risk individuals into family support systems to enhance overall stroke preparedness. Community nurses should implement family-centered health promotion activities, conduct regular home visits, and provide ongoing health education and guidance tailored to stroke prevention and early intervention. They should also educate family members about stroke symptoms and train high-risk patients to communicate effectively with their family members during symptom onset. Action plans to reduce prehospital delays should be developed, such as advising patients to immediately contact designated family members or friends when symptoms appear. (5) Focus on primary caregivers: Community workers should strengthen the stroke preparedness of primary caregivers by providing targeted education on stroke prevention, early symptom recognition, emergency response, and the urgency of timely treatment. Community lectures and health promotion activities should be organized to disseminate stroke-related knowledge, promote early diagnosis, encourage routine checkups, and enhance community-wide stroke preparedness. Stroke prevention and control efforts can be more effectively coordinated with the implementation of hospital-based graded diagnosis and treatment systems. This initiative can raise awareness, address misconceptions, and correct behaviors among primary caregivers and ultimately reduce prehospital delays for individuals at risk of stroke.

This study has some limitations. First, individuals who were unable to communicate effectively due to cognitive impairments or speech disorders at the time of the investigation were excluded, resulting in a sample that may not be representative of all patients with stroke. Second,

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the data were collected through a questionnaire survey, which may introduce subjective reporting bias. Our study has several limitations pertaining to bias and generalizability. First, outcomes were self-reported, introducing the potential for recall bias. Second, as data were collected only from consenting participants, we lacked information on eligible nonrespondents. This precludes assessment of nonresponse bias and may limit the generalizability of our findings to the broader stroke population. Third, by design, our enrollment was restricted to elderly patients without cognitive impairment to ensure reliable self-reporting. While methodologically necessary, this criterion inherently introduces selection bias by excluding frailer individuals who may have more severe strokes or additional comorbidities. Finally, we did not account for death as a competing risk, which may lead to an overestimation of cumulative incidences for nonfatal events, particularly among those with more severe strokes or delayed presentation. Further research is required to develop social network-based interventions aimed at preventing prehospital delays among high-risk stroke populations.

In conclusion, this study indicated that certain social network characteristics are significantly associated with prehospital delay in stroke care. Specifically, a larger effective network size, a higher percentage of kin, and the presence of individuals who immediately seek medical help when symptoms appear are protective factors against prehospital delay. In contrast, higher betweenness centrality, greater network constraints, and increased network efficiency are associated with an increased risk for prehospital delay.

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Disclosures

None.

Supplemental Methods

Supplemental File 1
Table S1
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